ABBA2 Implicit with Hairer's Symmetric Projection

Reduced physical derivation and canonical configuration space

Endpoint-time A–B–B–A map with symmetric diagonal projection

1 Scope and implementation name

The public ABBA family contains exactly five numerical-method classes:

- `ABBA2Midpoint`;
- `ABBA2Implicit`;
- `ABBA4Implicit`;
- `ABBA4ImplicitSingleProjection`;
- `ABBA6Implicit`.

State-space variants are parameters of these classes, not additional public method classes. The four implicit methods expose three independent axes:

Axis	Canonical values
`projection_formulation`	`reduced_multiplier`; `simultaneous_state_multiplier`
`nonlinear_solver`	`newton`; `broyden`
`state_extension`	`physical`; `shared_time`; `fully_extended`

Each implicit class therefore admits $2 \times 2 \times 3 = 12$ configurations. `ABBA2Midpoint` has no nonlinear residual and accepts only the three state extensions. The five public classes consequently represent $4 \times 12 + 3 = 51$ valid configurations.

The formulation and solver choices are global within a composed step. All three signed maps of `ABBA4Implicit`, and all seven signed maps of `ABBA6Implicit`, use the same configuration. In contrast, `ABBA4ImplicitSingleProjection` surrounds its complete unprojected triple jump with one projection and therefore performs one nonlinear solve per outer step.

This note derives the following branch of `ABBA2Implicit`:

- `projection_formulation`: `reduced_multiplier`;
- `state_extension`: `physical`.

Its nonlinear unknown is $\mu \in \mathbb{R}^2$ per particle and its exact Newton matrix is 2×2. The simultaneous formulation instead solves for $(u_f, v_f, \mu) \in \mathbb{R}^6$ per particle with a 6×6 Newton block. Both formulations define the same exact projected map when converged; Newton and Broyden are alternative algorithms for finding that root. The six-dimensional simultaneous vector is only an algebraic solver workspace, not a trajectory state or a separate numerical method.

2 Guiding-center system and duplicated phase space

The physical state is

$$z = (X, Y) \in \mathbb{R}^2,$$

and, with the convention used here, it satisfies

$$\dot{X} = -\partial_Y \phi(t, X, Y), \qquad \dot{Y} = \partial_X \phi(t, X, Y).$$

We define

$$J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad f(z,t) = J_0 \nabla_z \phi(z,t),$$

so that $\dot{z} = f(z,t)$. Its spatial Jacobian is

$$W(z,t) := D_z f(z,t) = J_0 \nabla_z^2 \phi(z,t).$$

Since the Hessian of ϕ is symmetric,

$$W(z,t)^T J_0 + J_0 W(z,t) = 0. \tag{1}$$

We duplicate the state:

$$\mathcal{Y} = (u,v) = (X_1, Y_1, X_2, Y_2) \in \mathbb{R}^4, \qquad u, v \in \mathbb{R}^2.$$

The duplicated dynamics associated with the subflows used below is

$$\dot{u} = f(v,t), \qquad \dot{v} = f(u,t).$$

If initially $u = v = z$, both equations coincide with the physical dynamics.

The physical state is identified with the diagonal manifold

$$\mathcal{M} = \{(u,v) \in \mathbb{R}^4 : u = v\}.$$

We write it as

$$g(\mathcal{Y}) = G\mathcal{Y} = u - v, \qquad G = (I_2 \ \ -I_2), \qquad GG^T = 2I_2.$$

For $\mu = (\mu_X, \mu_Y)^T$,

$$G^T \mu = \begin{pmatrix} \mu \\ -\mu \end{pmatrix} = (\mu_X, \mu_Y, -\mu_X, -\mu_Y)^T.$$

The constraints are linear. Therefore,

$$\nabla^2 g_1 = \nabla^2 g_2 = 0$$

and, for every μ,

$$\boxed{H_\mu(\mathcal{Y}) = \sum_{j=1}^{2} \mu_j \nabla^2 g_j(\mathcal{Y}) = 0.}$$

This does not mean that $\mu = 0$: H_μ is identically zero because $G = Dg$ is constant.

3 The explicit submaps A and B

For a substep length s and an evaluation time t, we define

$$A_{s,t}(u,v) = (u + sf(v,t), v), \tag{2}$$

$$B_{s,t}(u,v) = (u, v + sf(u,t)). \tag{3}$$

The map A advances only the first copy using the second one; B advances only the second copy using the first one. Both maps are explicit.

Their Jacobians are

$$DA_{s,t}(u,v) = \begin{pmatrix} I_2 & sW(v,t) \\ 0 & I_2 \end{pmatrix}, \qquad DB_{s,t}(u,v) = \begin{pmatrix} I_2 & 0 \\ sW(u,t) & I_2 \end{pmatrix}. \tag{4}$$

In the duplicated phase space, we use the skew-symmetric matrix

$$\Omega_\times = \begin{pmatrix} 0 & J_0 \\ J_0 & 0 \end{pmatrix}.$$

Identity (1) directly implies

$$(DA_{s,t})^T \Omega_\times DA_{s,t} = \Omega_\times, \qquad (DB_{s,t})^T \Omega_\times DB_{s,t} = \Omega_\times.$$

Thus, each submap is symplectic with respect to $\Omega_\times$, and any composition of these maps is also symplectic in the duplicated phase space.

Moreover, the variables used to evaluate the field remain fixed in each submap. Therefore,

$$\boxed{A_{s,t}^{-1} = A_{-s,t}, \qquad B_{s,t}^{-1} = B_{-s,t}.} \tag{5}$$

4 The nonautonomous A–B–B–A step

Let

$$t_{n+1} = t_n + h, \qquad s = \frac{h}{2}.$$

The base map that replaces BM4 is

$$\boxed{\Phi_{h,t_n}^{\mathrm{ABBA}} = A_{s,t_{n+1}} \circ B_{s,t_{n+1}} \circ B_{s,t_n} \circ A_{s,t_n}.} \tag{6}$$

Compositions are read from right to left. Therefore, the effective order is

$$A_{s,t_n} \longrightarrow B_{s,t_n} \longrightarrow B_{s,t_{n+1}} \longrightarrow A_{s,t_{n+1}}.$$

Starting from $(u^{(0)}, v^{(0)})$, the four stages are

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \qquad (A_1) \tag{7a}$$

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \qquad (B_1) \tag{7b}$$

$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \qquad (B_2) \tag{7c}$$

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}). \qquad (A_2) \tag{7d}$$

The output is

$$\Phi_{h,t_n}^{\mathrm{ABBA}}(u^{(0)}, v^{(0)}) = (u^{(2)}, v^{(2)}).$$

The evaluations at the two time endpoints are essential in the nonautonomous problem. If every stage were evaluated at t_n, time symmetry would be lost. In the autonomous case, the two central stages can be combined, and one recovers

$$A_{h/2} \longrightarrow B_h \longrightarrow A_{h/2}.$$

Since A and B are symplectic with respect to $\Omega_\times$, the complete map is also symplectic:

$$\boxed{(D\Phi_{h,t_n}^{\mathrm{ABBA}})^T \Omega_\times D\Phi_{h,t_n}^{\mathrm{ABBA}} = \Omega_\times.} \tag{8}$$

5 Proof of the symmetry of ABBA

A time-dependent method is symmetric if

$$\Phi^{\mathrm{ABBA}}_{-h,t_{n+1}} = \left(\Phi^{\mathrm{ABBA}}_{h,t_n}\right)^{-1}. \tag{9}$$

Using (5) and reversing the order of composition in (6),

$$\left(\Phi^{\mathrm{ABBA}}_{h,t_n}\right)^{-1} = A_{-s,t_n} \circ B_{-s,t_n} \circ B_{-s,t_{n+1}} \circ A_{-s,t_{n+1}}.$$

We now start at t_{n+1} with a step of size $-h$. Its final time is $t_{n+1} - h = t_n$, and definition (6) gives precisely

$$\Phi^{\mathrm{ABBA}}_{-h,t_{n+1}} = A_{-s,t_n} \circ B_{-s,t_n} \circ B_{-s,t_{n+1}} \circ A_{-s,t_{n+1}}.$$

Therefore,

$$\boxed{\Phi^{\mathrm{ABBA}}_{-h,t_{n+1}} = \left(\Phi^{\mathrm{ABBA}}_{h,t_n}\right)^{-1}.}$$

The architecture is palindromic when both the sign of the step and the two endpoint times are reversed simultaneously.

6 Hairer's symmetric projection around ABBA

Suppose that

$$\mathcal{Y}_n = (z_n, z_n) \in \mathcal{M}$$

is known. For a multiplier $\mu \in \mathbb{R}^2$, we first correct the input:

$$\widehat{\mathcal{Y}}_n(\mu) = \mathcal{Y}_n + G^T\mu = (z_n + \mu, z_n - \mu).$$

We write

$$u^{(0)} = z_n + \mu, \qquad v^{(0)} = z_n - \mu,$$

and apply the four stages in (7). The auxiliary result is

$$\widetilde{\mathcal{Y}}_{n+1}(\mu) = \Phi^{\mathrm{ABBA}}_{h,t_n}(\mathcal{Y}_n + G^T\mu) = (u^{(2)}(\mu), v^{(2)}(\mu)).$$

The same correction is then applied at the output:

$$\mathcal{Y}_{n+1}(\mu) = \widetilde{\mathcal{Y}}_{n+1}(\mu) + G^T\mu = (u^{(2)} + \mu, v^{(2)} - \mu). \tag{10}$$

Imposing $\mathcal{Y}_{n+1} \in \mathcal{M}$ is equivalent to

$$u^{(2)}(\mu) + \mu = v^{(2)}(\mu) - \mu.$$

Thus, the problem is reduced exactly to two equations:

$$\boxed{R(\mu) := u^{(2)}(\mu) - v^{(2)}(\mu) + 2\mu = 0.} \tag{11}$$

In matrix notation,

$$R(\mu) = G\Phi^{\mathrm{ABBA}}_{h,t_n}(\mathcal{Y}_n + G^T\mu) + 2\mu. \tag{12}$$

Once the root μ_* has been found, the new physical state is

$$\boxed{z_{n+1} = u^{(2)}(\mu_*) + \mu_* = v^{(2)}(\mu_*) - \mu_*.} \tag{13}$$

7 Exact Jacobian of the ABBA map

For a particular evaluation of (7), we define

$$W_1 = W(v^{(0)}, t_n), \qquad W_2 = W(u^{(1)}, t_n),$$
$$W_3 = W(u^{(1)}, t_{n+1}), \qquad W_4 = W(v^{(2)}, t_{n+1}),$$

and abbreviate

$$S := W_2 + W_3.$$

Explicit expansion of $u^{(2)}$

To compute the derivative of the last stage without hiding any dependence, we start from

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1})$$

and successively substitute the preceding stages. First,

$$\begin{aligned} v^{(2)} &= v^{(1)} + sf(u^{(1)}, t_{n+1}) \\ &= v^{(0)} + sf(u^{(1)}, t_n) + sf(u^{(1)}, t_{n+1}). \end{aligned}$$

Since

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n),$$

the expression for $v^{(2)}$ solely in terms of the initial data is

$$\begin{aligned} v^{(2)} = v^{(0)} &+ sf\Big(u^{(0)} + sf(v^{(0)}, t_n), t_n\Big) \\ &+ sf\Big(u^{(0)} + sf(v^{(0)}, t_n), t_{n+1}\Big). \end{aligned}$$

Therefore, complete substitution into the last stage gives

$$\boxed{\begin{aligned} u^{(2)} = u^{(0)} &+ sf(v^{(0)}, t_n) \\ &+ sf\Big(v^{(0)} + sf\Big(u^{(0)} + sf(v^{(0)}, t_n), t_n\Big) \\ &\qquad + sf\Big(u^{(0)} + sf(v^{(0)}, t_n), t_{n+1}\Big), t_{n+1}\Big). \end{aligned}}$$

This formula displays directly all dependencies of $u^{(2)}$ on $u^{(0)}$ and $v^{(0)}$. Nevertheless, to apply the chain rule readably, it is convenient to differentiate the intermediate stages in the same order in which they are computed.

For the first stage, we have

$$\frac{\partial u^{(1)}}{\partial u^{(0)}} = I_2, \qquad \frac{\partial u^{(1)}}{\partial v^{(0)}} = sW_1.$$

Differentiating the expression for $v^{(2)}$ yields

$$\begin{aligned} \frac{\partial v^{(2)}}{\partial u^{(0)}} &= sW_2 \frac{\partial u^{(1)}}{\partial u^{(0)}} + sW_3 \frac{\partial u^{(1)}}{\partial u^{(0)}} = s(W_2 + W_3) = sS, \\ \frac{\partial v^{(2)}}{\partial v^{(0)}} &= I_2 + sW_2 \frac{\partial u^{(1)}}{\partial v^{(0)}} + sW_3 \frac{\partial u^{(1)}}{\partial v^{(0)}} \\ &= I_2 + s^2(W_2 + W_3)W_1 = I_2 + s^2 S W_1. \end{aligned}$$

Finally, differentiating $u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1})$, we obtain

$$\frac{\partial u^{(2)}}{\partial u^{(0)}} = \frac{\partial u^{(1)}}{\partial u^{(0)}} + sW_4 \frac{\partial v^{(2)}}{\partial u^{(0)}}$$

$$
\begin{aligned}
&= I_2 + s^2 W_4 S, \\
\frac{\partial u^{(2)}}{\partial v^{(0)}} &= \frac{\partial u^{(1)}}{\partial v^{(0)}} + sW_4 \frac{\partial v^{(2)}}{\partial v^{(0)}} \\
&= sW_1 + sW_4(I_2 + s^2 SW_1) \\
&= s(W_1 + W_4) + s^3 W_4 S W_1.
\end{aligned}
$$

Consequently, the two blocks in the first row of the ABBA Jacobian are

$$
\boxed{A := \frac{\partial u^{(2)}}{\partial u^{(0)}} = I_2 + s^2 W_4 S, \qquad B := \frac{\partial u^{(2)}}{\partial v^{(0)}} = s(W_1 + W_4) + s^3 W_4 S W_1.}
$$

The order of the matrix products follows from the chain rule and must be preserved because the matrices W_i evaluated at different points need not commute.

The Jacobians of the four stages, evaluated at the points actually traversed, are

$$
M_{A_1} = \begin{pmatrix} I_2 & sW_1 \\ 0 & I_2 \end{pmatrix}, \qquad M_{B_1} = \begin{pmatrix} I_2 & 0 \\ sW_2 & I_2 \end{pmatrix},
$$

$$
M_{B_2} = \begin{pmatrix} I_2 & 0 \\ sW_3 & I_2 \end{pmatrix}, \qquad M_{A_2} = \begin{pmatrix} I_2 & sW_4 \\ 0 & I_2 \end{pmatrix}.
$$

By the chain rule,

$$
\boxed{D\Phi_{h,t_n}^{\mathrm{ABBA}} = M_{A_2} M_{B_2} M_{B_1} M_{A_1}.} \tag{14}
$$

The two central matrices satisfy

$$
M_{B_2} M_{B_1} = \begin{pmatrix} I_2 & 0 \\ sS & I_2 \end{pmatrix}.
$$

Multiplying the four blocks gives the exact 4×4 Jacobian:

$$
\boxed{D\Phi_{h,t_n}^{\mathrm{ABBA}} = \begin{pmatrix} I_2 + s^2 W_4 S & s(W_1 + W_4) + s^3 W_4 S W_1 \\ sS & I_2 + s^2 S W_1 \end{pmatrix}.} \tag{15}
$$

The Jacobian of the map has not been replaced by the identity: (15) contains exactly all dependencies generated by the four stages.

8 Exact Jacobian of the reduced residual

From

$$
\frac{\partial(u^{(0)}, v^{(0)})}{\partial \mu} = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix} = G^T
$$

and from (12), we obtain

$$
\boxed{R'(\mu) = G\, D\Phi_{h,t_n}^{\mathrm{ABBA}}(\mathcal{Y}_n + G^T \mu) G^T + 2I_2.} \tag{16}
$$

If we write (15) as

$$
D\Phi_{h,t_n}^{\mathrm{ABBA}} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},
$$

then

$$
G\, D\Phi_{h,t_n}^{\mathrm{ABBA}}\, G^T = A - B - C + D.
$$

Substituting the four blocks in (15) yields

$$\boxed{\begin{aligned} R'(\mu) = 4I_2 &- s(W_1+W_2+W_3+W_4) \\ &+ s^2\big[W_4(W_2+W_3)+(W_2+W_3)W_1\big] \\ &- s^3 W_4(W_2+W_3)W_1. \end{aligned}} \tag{17}$$

Since $s = h/2$, this can also be written as

$$\boxed{\begin{aligned} R'(\mu) = 4I_2 &- \frac{h}{2}(W_1+W_2+W_3+W_4) \\ &+ \frac{h^2}{4}\big[W_4(W_2+W_3)+(W_2+W_3)W_1\big] \\ &- \frac{h^3}{8} W_4(W_2+W_3)W_1. \end{aligned}} \tag{18}$$

The matrices W_i need not commute: the order of the products in (17) and (18) must be preserved exactly.

9 Explicit Jacobian of the implicit physical step

The Jacobian $R'(\mu)$ computed in the previous section is the matrix used to solve the nonlinear system that determines the multiplier. However, it is not the Jacobian of the physical numerical method. The object needed, for example, to verify the symplecticity of the step directly is

$$D\Psi_{h,t_n}(z_n) = \frac{\partial z_{n+1}}{\partial z_n} \in \mathbb{R}^{2\times 2},$$

where

$$\Psi_{h,t_n} : z_n \longmapsto z_{n+1}.$$

The difference between these matrices is essential: changing z_n also changes the converged root $\mu_* = \mu_h(z_n)$. Therefore, computing $D\Psi_{h,t_n}(z_n)$ requires implicit differentiation of the multiplier. Throughout this section, the matrices W_i and the Jacobian $D\Phi^{\mathrm{ABBA}}_{h,t_n}$ are evaluated along the ABBA stages corresponding to the converged solution $\mu_h(z_n)$. In particular, for GC2D,

$$W_i = \begin{pmatrix} -\phi_{XY} & -\phi_{YY} \\ \phi_{XX} & \phi_{XY} \end{pmatrix}_{(z_i,t_i)}.$$

The four points (z_i, t_i) are, respectively, $(v^{(0)}, t_n)$, $(u^{(1)}, t_n)$, $(u^{(1)}, t_{n+1})$, and $(v^{(2)}, t_{n+1})$.

9.1 Partial derivatives of the residual

We introduce the matrices

$$E = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix}, \qquad N = G^T = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix}.$$

The ABBA input state can be written as

$$\begin{pmatrix} u^{(0)} \\ v^{(0)} \end{pmatrix} = Ez_n + N\mu = \begin{pmatrix} z_n + \mu \\ z_n - \mu \end{pmatrix}.$$

To display the dependence on the initial state, we write the residual as a function of its two variables:

$$\mathcal{R}(z,\mu) = G\Phi^{\mathrm{ABBA}}_{h,t_n}(Ez + N\mu) + 2\mu.$$

Let
$$\mathcal{J}_{\mathrm{ABBA}} := D\Phi^{\mathrm{ABBA}}_{h,t_n} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$
with the blocks from (15), namely,
$$\begin{aligned} A &= I_2 + s^2 W_4 S, \quad & B &= s(W_1 + W_4) + s^3 W_4 S W_1, \\ C &= sS, & D &= I_2 + s^2 S W_1. \end{aligned}$$
When taking a partial derivative with respect to μ, the state z is held fixed. Since
$$\frac{\partial(Ez + N\mu)}{\partial\mu} = N,$$
the chain rule gives
$$\boxed{\mathcal{K} := D_\mu \mathcal{R} = G\mathcal{J}_{\mathrm{ABBA}} N + 2I_2.}$$
Similarly, when differentiating with respect to z, μ is held fixed and
$$\frac{\partial(Ez + N\mu)}{\partial z} = E.$$
Therefore,
$$\boxed{\mathcal{L} := D_z \mathcal{R} = G\mathcal{J}_{\mathrm{ABBA}} E.}$$
In terms of the four blocks,
$$\boxed{\mathcal{K} = A - B - C + D + 2I_2, \qquad \mathcal{L} = A + B - C - D.}$$
Substituting the explicit expressions for A, B, C, and D gives
$$\boxed{\begin{aligned} \mathcal{K} = 4I_2 &- s(W_1 + W_2 + W_3 + W_4) \\ &+ s^2[W_4 S + S W_1] - s^3 W_4 S W_1, \end{aligned}}$$
y
$$\boxed{\begin{aligned} \mathcal{L} = s&(W_1 - W_2 - W_3 + W_4) \\ &+ s^2[W_4 S - S W_1] + s^3 W_4 S W_1. \end{aligned}}$$
The matrix $\mathcal{K}$ coincides with $R'(\mu)$ from (17), evaluated at the root. In contrast, $\mathcal{L}$ measures how the residual changes when the initial physical state is perturbed while the multiplier is momentarily held fixed. All matrices in these two expressions are 2×2.

No derivatives of W_i appear in these formulas. This is because $\mathcal{J}_{\mathrm{ABBA}}$ is already the complete first derivative of the ABBA composition: each $W_i = D_z f(z_i, t_i)$ is evaluated at its corresponding stage point. Differentiating the W_i again would produce second derivatives of the map and is not part of the Jacobian sought here.

9.2 Implicit derivative of the multiplier

The converged multiplier satisfies
$$\mathcal{R}(z, \mu_h(z)) = 0.$$
Differentiating this identity with respect to z and applying the chain rule,
$$D_z\mathcal{R} + D_\mu\mathcal{R}\, D\mu_h(z) = 0.$$
With the notation above,
$$\mathcal{L} + \mathcal{K} D\mu_h(z) = 0.$$
If $\mathcal{K}$ is invertible, the implicit function theorem locally guarantees that μ_h is differentiable and
$$\boxed{D\mu_h(z) = -\mathcal{K}^{-1}\mathcal{L}.}$$
This term is exactly the contribution that would be lost if μ were treated as a constant independent of z when differentiating the step.

9.3 Differentiation of the final physical state

We use the first of the two equivalent expressions in (13):

$$z_{n+1} = u^{(2)} + \mu_h(z_n).$$

Como

$$u^{(0)} = z_n + \mu_h(z_n), \qquad v^{(0)} = z_n - \mu_h(z_n),$$

their derivatives with respect to z_n are

$$\frac{\partial u^{(0)}}{\partial z_n} = I_2 + D\mu_h(z_n), \qquad \frac{\partial v^{(0)}}{\partial z_n} = I_2 - D\mu_h(z_n).$$

To make the chain rule explicit, we introduce the notation

$$M_\mu := D\mu_h(z_n), \qquad U_j := \frac{\partial u^{(j)}}{\partial z_n}, \qquad V_j := \frac{\partial v^{(j)}}{\partial z_n}.$$

Thus, the two initial derivatives are

$$U_0 = I_2 + M_\mu, \qquad V_0 = I_2 - M_\mu.$$

We now differentiate the four ABBA stages directly with respect to z_n. In the first stage,

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n),$$

and the chain rule gives

$$U_1 = U_0 + sW_1V_0.$$

For the two central stages,

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \qquad v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}),$$

we obtain

$$\begin{aligned} V_1 &= V_0 + sW_2U_1, \\ V_2 &= V_1 + sW_3U_1 = V_0 + s(W_2 + W_3)U_1 = V_0 + sSU_1. \end{aligned}$$

Finally, the last stage is

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}),$$

so that

$$U_2 = U_1 + sW_4V_2.$$

We now substitute the expressions for U_1 and V_2 into that for U_2, without omitting any steps:

$$\begin{aligned} U_2 &= U_1 + sW_4(V_0 + sSU_1) \\ &= (I_2 + s^2W_4S)U_1 + sW_4V_0 \\ &= (I_2 + s^2W_4S)(U_0 + sW_1V_0) + sW_4V_0 \\ &= (I_2 + s^2W_4S)U_0 + \Big[sW_1 + sW_4 + s^3W_4SW_1\Big]V_0 \\ &= \underbrace{(I_2 + s^2W_4S)}_{A}\,U_0 + \underbrace{\Big[s(W_1 + W_4) + s^3W_4SW_1\Big]}_{B}\,V_0. \end{aligned}$$

The blocks computed previously appear here exactly,

$$A = \frac{\partial u^{(2)}}{\partial u^{(0)}}, \qquad B = \frac{\partial u^{(2)}}{\partial v^{(0)}}.$$

Therefore, substituting $U_0 = I_2 + M_\mu$ and $V_0 = I_2 - M_\mu$ explicitly gives

$$\begin{aligned}\frac{\partial u^{(2)}}{\partial z_n} &= A\big(I_2 + M_\mu\big) + B\big(I_2 - M_\mu\big)\\ &= A + AM_\mu + B - BM_\mu\\ &= A + B + (A - B)M_\mu\\ &= A + B + (A - B)D\mu_h(z_n).\end{aligned}$$

If the expressions for A and B are also substituted, the unabbreviated result is

$$\begin{aligned}\frac{\partial u^{(2)}}{\partial z_n} = I_2 &+ s(W_1 + W_4) + s^2 W_4 S + s^3 W_4 S W_1\\ &+ \Big[I_2 - s(W_1 + W_4) + s^2 W_4 S - s^3 W_4 S W_1\Big] D\mu_h(z_n).\end{aligned}$$

The order of the products matters: $D\mu_h(z_n)$ appears on the right because Jacobians act on perturbations as matrices multiplying from the left.

Differentiating the final correction $+\mu_h(z_n)$ as well gives

$$\boxed{D\Psi_{h,t_n}(z_n) = A + B + (A - B + I_2)D\mu_h(z_n).}$$

Finally, substituting the implicit derivative of the multiplier,

$$\boxed{D\Psi_{h,t_n}(z_n) = A + B - (A - B + I_2)\mathcal{K}^{-1}\mathcal{L}.}$$

To abbreviate this expression, we define

$$\mathcal{P} := A + B, \qquad \mathcal{Q} := A - B + I_2.$$

Their explicit values are

$$\boxed{\mathcal{P} = I_2 + s(W_1 + W_4) + s^2 W_4 S + s^3 W_4 S W_1,}$$

$$\boxed{\mathcal{Q} = 2I_2 - s(W_1 + W_4) + s^2 W_4 S - s^3 W_4 S W_1.}$$

Consequently, the exact Jacobian of the implicit physical step is given by the compact formula

$$\boxed{D\Psi_{h,t_n}(z_n) = \mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L}.}$$

This formula includes both the direct variation of the four ABBA stages and the indirect variation due to the dependence $\mu_h = \mu_h(z_n)$.

9.4 Entrywise expression

Since all the preceding matrices are 2×2, the only required inverse can be written in closed form. Let

$$\mathcal{P} = (p_{ij}), \qquad \mathcal{Q} = (q_{ij}), \qquad \mathcal{K} = \begin{pmatrix} k_{11} & k_{12}\\ k_{21} & k_{22}\end{pmatrix}, \qquad \mathcal{L} = \begin{pmatrix} \ell_{11} & \ell_{12}\\ \ell_{21} & \ell_{22}\end{pmatrix}.$$

Si

$$\Delta_{\mathcal{K}} := \det \mathcal{K} = k_{11}k_{22} - k_{12}k_{21} \neq 0,$$

then

$$\mathcal{K}^{-1}\mathcal{L} = \frac{1}{\Delta_{\mathcal{K}}}\begin{pmatrix} k_{22}\ell_{11} - k_{12}\ell_{21} & k_{22}\ell_{12} - k_{12}\ell_{22}\\ -k_{21}\ell_{11} + k_{11}\ell_{21} & -k_{21}\ell_{12} + k_{11}\ell_{22}\end{pmatrix}.$$

Let

$$\mathcal{T} := \mathcal{K}^{-1}\mathcal{L} = \begin{pmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{pmatrix}.$$

The physical Jacobian is then

$$\boxed{D\Psi_{h,t_n} = \begin{pmatrix} p_{11} - q_{11}\tau_{11} - q_{12}\tau_{21} & p_{12} - q_{11}\tau_{12} - q_{12}\tau_{22} \\ p_{21} - q_{21}\tau_{11} - q_{22}\tau_{21} & p_{22} - q_{21}\tau_{12} - q_{22}\tau_{22} \end{pmatrix}}.$$

That is,

$$D\Psi_{h,t_n}(z_n) = \begin{pmatrix} \dfrac{\partial X_{n+1}}{\partial X_n} & \dfrac{\partial X_{n+1}}{\partial Y_n} \\ \dfrac{\partial Y_{n+1}}{\partial X_n} & \dfrac{\partial Y_{n+1}}{\partial Y_n} \end{pmatrix}.$$

As a check, for $h = 0$ we have $s = 0$ and therefore

$$\mathcal{K} = 4I_2, \qquad \mathcal{L} = 0, \qquad \mathcal{P} = I_2, \qquad \mathcal{Q} = 2I_2.$$

The preceding formula gives $D\Psi_{0,t_n} = I_2$, as required for the identity map.
It is this physical matrix, rather than $R'(\mu) = \mathcal{K}$, that must be used to verify directly

$$(D\Psi_{h,t_n})^T J_0 D\Psi_{h,t_n} = J_0,$$

or, equivalently in two dimensions,

$$\det D\Psi_{h,t_n} = 1.$$

10 Symmetry of the complete projected method

The equation defining the projected step can be written as

$$\boxed{\mathcal{Y}_{n+1} - G^T\mu = \Phi^{\text{ABBA}}_{h,t_n}(\mathcal{Y}_n + G^T\mu), \qquad \mathcal{Y}_n, \mathcal{Y}_{n+1} \in \mathcal{M}.} \tag{19}$$

Applying the inverse of ABBA and using its symmetry,

$$\mathcal{Y}_n + G^T\mu = \Phi^{\text{ABBA}}_{-h,t_{n+1}}(\mathcal{Y}_{n+1} - G^T\mu).$$

We define the multiplier of the reverse step as

$$\mu_{\text{inv}} = -\mu.$$

Then

$$\mathcal{Y}_n - G^T\mu_{\text{inv}} = \Phi^{\text{ABBA}}_{-h,t_{n+1}}(\mathcal{Y}_{n+1} + G^T\mu_{\text{inv}}),$$

which is exactly equation (19) applied backward. Therefore, if the root is locally unique and is solved exactly,

$$\boxed{\Psi_{-h,t_{n+1}} \circ \Psi_{h,t_n} = I.} \tag{20}$$

Here Ψ denotes ABBA together with Hairer's projection. In an implementation, the iteration is stopped at a finite tolerance; reversibility is preserved up to that tolerance.

11 Symplecticity of the complete projected method

We now prove that the physical map obtained after symmetric projection is symplectic. The point requiring care is that the multiplier is not constant: although it is found iteratively within each time step, its converged value depends implicitly on the initial physical state z_n.

11.1 Diagonal and correction directions

Recall the matrices introduced above

$$E = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix}, \qquad N = G^T = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix}.$$

Thus, $Ez = (z, z)$ belongs to the diagonal manifold, while $N\mu = (\mu, -\mu)$ gives the correction direction. If

$$z_{n+1} = \Psi_{h,t_n}(z_n),$$

then equation (19) is equivalent to

$$\boxed{\Phi^{\mathrm{ABBA}}_{h,t_n}(Ez + N\mu_h(z)) = E\Psi_{h,t_n}(z) - N\mu_h(z).} \tag{21}$$

The same multiplier is used on both sides of this equation within one time step, with opposite signs. This does not mean that the multiplier remains unchanged in the next step.

11.2 The residual and the implicit dependence of the multiplier

In (11)–(12), z_n was fixed and the residual was therefore written only as a function of μ. To study derivatives of the complete physical map, we restore this dependence and write

$$\boxed{\mathcal{R}(z, \mu) := G\Phi^{\mathrm{ABBA}}_{h,t_n}(Ez + N\mu) + 2\mu.} \tag{22}$$

Detailed construction of equation (22)

Expression (22) combines in a single function the two conditions required for the two state copies to coincide again after applying ABBA and the symmetric correction. We now reconstruct it step by step.

We start from a physical state $z \in \mathbb{R}^2$. The matrix E embeds it in the diagonal manifold of the duplicated space:

$$Ez = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix} z = \begin{pmatrix} z \\ z \end{pmatrix} = (z, z) \in \mathbb{R}^4.$$

The matrix $N = G^T$ introduces the multiplier in the direction transverse to that manifold:

$$N\mu = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix} \mu = \begin{pmatrix} \mu \\ -\mu \end{pmatrix} = (\mu, -\mu) \in \mathbb{R}^4.$$

Therefore, the ABBA argument in (22) is the pre-corrected state

$$Ez + N\mu = (z + \mu, z - \mu).$$

The two copies are separated before integration by equal corrections with opposite signs. This separation is intentional: μ is chosen so that the complete symmetric correction makes both copies coincide again at the end of the step.

Applying the ABBA map gives an auxiliary state in the duplicated space:

$$\Phi^{\mathrm{ABBA}}_{h,t_n}(Ez + N\mu) = \begin{pmatrix} u^{(2)}(z, \mu) \\ v^{(2)}(z, \mu) \end{pmatrix}.$$

Here the dependence on z and μ has been written explicitly. In general, the four ABBA stages give $u^{(2)}(z, \mu) \neq v^{(2)}(z, \mu)$.

The matrix

$$G = (I_2 \ \ -I_2) : \mathbb{R}^4 \longrightarrow \mathbb{R}^2$$

measures exactly this mismatch, since for every $(u, v) \in \mathbb{R}^4$,

$$G \begin{pmatrix} u \\ v \end{pmatrix} = u - v.$$

In particular,

$$G\Phi_{h,t_n}^{\mathrm{ABBA}}(Ez + N\mu) = u^{(2)}(z, \mu) - v^{(2)}(z, \mu).$$

This term is the diagonal-constraint defect before applying the output correction.

The symmetric projection uses the same multiplier at the output. The final duplicated state is

$$\mathcal{Y}_{n+1} = \Phi_{h,t_n}^{\mathrm{ABBA}}(Ez + N\mu) + N\mu.$$

To require this state to belong to $\mathcal{M}$, we impose

$$G\mathcal{Y}_{n+1} = 0.$$

Substituting the preceding expression,

$$G\mathcal{Y}_{n+1} = G\Phi_{h,t_n}^{\mathrm{ABBA}}(Ez + N\mu) + GN\mu.$$

Since $N = G^T$ and $GG^T = 2I_2$, we obtain

$$GN\mu = GG^T\mu = 2\mu.$$

This calculation explains the term 2μ in (22): it is neither an approximation nor an arbitrary factor, but the exact contribution of the output correction to the constraint defect. Consequently,

$$G\mathcal{Y}_{n+1} = G\Phi_{h,t_n}^{\mathrm{ABBA}}(Ez + N\mu) + 2\mu = \mathcal{R}(z, \mu).$$

Therefore, the following statements are equivalent:

$$\boxed{\mathcal{R}(z, \mu) = 0 \quad \Longleftrightarrow \quad G\mathcal{Y}_{n+1} = 0 \quad \Longleftrightarrow \quad \mathcal{Y}_{n+1} \in \mathcal{M}.}$$

Writing the equality componentwise, with $u^{(2)} = (u_X^{(2)}, u_Y^{(2)})^T$, $v^{(2)} = (v_X^{(2)}, v_Y^{(2)})^T$, and $\mu = (\mu_X, \mu_Y)^T$, the nonlinear system (22) is

$$\mathcal{R}(z, \mu) = \begin{pmatrix} u_X^{(2)}(z, \mu) - v_X^{(2)}(z, \mu) + 2\mu_X \\ u_Y^{(2)}(z, \mu) - v_Y^{(2)}(z, \mu) + 2\mu_Y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus, two equations are solved for the two components of μ. Four equations are not needed: membership in the diagonal only requires the difference between the copies to vanish.

It is also useful to check the dimensions of each object:

$$z, \mu \in \mathbb{R}^2, \qquad E, N \in \mathbb{R}^{4\times 2}, \qquad Ez + N\mu \in \mathbb{R}^4,$$
$$\Phi_{h,t_n}^{\mathrm{ABBA}} : \mathbb{R}^4 \to \mathbb{R}^4, \qquad G \in \mathbb{R}^{2\times 4}, \qquad \mathcal{R}(z, \mu) \in \mathbb{R}^2.$$

Thus, (22) correctly defines a map

$$\mathcal{R} : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2.$$

When z is fixed during a time step, one may abbreviate $R(\mu) = \mathcal{R}(z, \mu)$, as in (11)–(12). However, differentiating the complete physical map requires retaining the dependence on z, because changing the initial state changes all four ABBA stages and hence the root of the projection system.

As a simple check, if $h = 0$, then $\Phi^{\mathrm{ABBA}}_{h,t_n} = I_4$. In that case,

$$\mathcal{R}(z,\mu) = G(Ez + N\mu) + 2\mu = GEz + GN\mu + 2\mu.$$

Because $GE = 0$ and $GN = 2I_2$, it follows that

$$\mathcal{R}(z,\mu) = 4\mu.$$

The unique root is $\mu = 0$, as required when no time advance occurs. Moreover, $D_\mu\mathcal{R} = 4I_2$ for $h = 0$; by continuity, this matrix generally remains invertible for sufficiently small steps, allowing local use of the implicit function theorem.

The equation determining the multiplier is

$$\mathcal{R}(z,\mu) = 0.$$

Consequently, the multiplier obtained after convergence can be written locally as

$$\mu = \mu_h(z).$$

This does not assert that an explicit formula for μ_h is available. It only states that changing z changes the nonlinear system and therefore its converged solution.

The Jacobian of the residual with respect to the multiplier is

$$D_\mu\mathcal{R} = G\,D\Phi^{\mathrm{ABBA}}_{h,t_n}\,N + 2I_2. \tag{23}$$

This is precisely the matrix $R'(\mu)$ obtained in (16). If it is invertible at the solution, the implicit function theorem guarantees that $\mu_h(z)$ is locally differentiable. In fact,

$$D\mu_h(z) = -\left(G\,D\Phi^{\mathrm{ABBA}}_{h,t_n}\,N + 2I_2\right)^{-1} G\,D\Phi^{\mathrm{ABBA}}_{h,t_n}\,E,$$

where $D\Phi^{\mathrm{ABBA}}_{h,t_n}$ is evaluated at $Ez + N\mu_h(z)$.

Define

$$M := D\mu_h(z), \qquad J_\Psi := D\Psi_{h,t_n}(z).$$

Differentiating (21) with respect to z gives

$$\boxed{D\Phi^{\mathrm{ABBA}}_{h,t_n}\,(E + NM) = EJ_\Psi - NM.} \tag{24}$$

For a physical perturbation $u \in \mathbb{R}^2$, this identity states that

$$(E + NM)u \xrightarrow{D\Phi^{\mathrm{ABBA}}_{h,t_n}} (EJ_\Psi - NM)u.$$

Here $Mu = D\mu_h(z)u$ compares converged multipliers associated with nearby initial states; it is not the difference between two internal iterations.

11.3 Symplectic orthogonality of the two directions

Direct multiplication using the definition of $\Omega_\times$ gives

$$\boxed{\begin{aligned} &E^T\Omega_\times E = 2J_0, \quad N^T\Omega_\times N = -2J_0,\\ &E^T\Omega_\times N = 0, \qquad N^T\Omega_\times E = 0. \end{aligned}} \tag{25}$$

Therefore, the diagonal direction (u, u) and correction direction $(a, -a)$ are symplectically orthogonal with respect to $\Omega_\times$.

11.4 Conservation of the extended symplectic form

Equation (24) and the symplecticity of ABBA in (8) imply

$$\begin{aligned}
&(EJ_\Psi - NM)^T\Omega_\times(EJ_\Psi - NM)\\
&\quad= \left[D\Phi^{\mathrm{ABBA}}_{h,t_n}(E+NM)\right]^T\Omega_\times\left[D\Phi^{\mathrm{ABBA}}_{h,t_n}(E+NM)\right]\\
&\quad= (E+NM)^T(D\Phi^{\mathrm{ABBA}}_{h,t_n})^T\Omega_\times D\Phi^{\mathrm{ABBA}}_{h,t_n}(E+NM)\\
&\quad= (E+NM)^T\Omega_\times(E+NM).
\end{aligned} \tag{26}$$

The last equality is exactly where symplecticity of the duplicated ABBA map is used. Thus, (26) states that $D\Phi^{\mathrm{ABBA}}_{h,t_n}$ preserves $\Omega_\times$ between the initial and final lifted perturbations.

Using the four identities in (25), the initial side becomes

$$\begin{aligned}
(E+NM)^T\Omega_\times(E+NM) &= E^T\Omega_\times E + M^TN^T\Omega_\times NM\\
&= 2J_0 - 2M^TJ_0M.
\end{aligned} \tag{27}$$

The mixed terms vanish because $E^T\Omega_\times N = N^T\Omega_\times E = 0$. Likewise, the final side is

$$\begin{aligned}
(EJ_\Psi - NM)^T\Omega_\times(EJ_\Psi - NM) &= J_\Psi^TE^T\Omega_\times EJ_\Psi + M^TN^T\Omega_\times NM\\
&= 2J_\Psi^TJ_0J_\Psi - 2M^TJ_0M.
\end{aligned} \tag{28}$$

Substituting (27) and (28) into (26) gives

$$2J_\Psi^TJ_0J_\Psi - 2M^TJ_0M = 2J_0 - 2M^TJ_0M. \tag{29}$$

The term produced by the implicit dependence of the multiplier is identical on both sides and cancels. Hence,

$$\boxed{J_\Psi^TJ_0J_\Psi = J_0.} \tag{30}$$

Since $J_\Psi = D\Psi_{h,t_n}(z)$,

$$\boxed{D\Psi_{h,t_n}(z)^TJ_0D\Psi_{h,t_n}(z) = J_0.} \tag{31}$$

Therefore, the complete physical method, consisting of ABBA together with Hairer's symmetric projection, is symplectic on the physical phase space. In two dimensions this also gives

$$\det D\Psi_{h,t_n}(z) = 1,$$

so the method preserves oriented area in the (X, Y) plane.

The proof requires the following local hypotheses:

1. the ABBA map is symplectic with respect to $\Omega_\times$;
2. the equation $\mathcal{R}(z,\mu) = 0$ has a locally unique solution;
3. $D_\mu\mathcal{R}$ is invertible, so $\mu_h(z)$ is differentiable;
4. the nonlinear projection equation is solved exactly.

The reduced multiplier system defines the same exact projected map independently of the particular convergent nonlinear solver. With a finite stopping tolerance, symplecticity is preserved only up to the defect introduced by that tolerance.

12 State extensions and dimensional conventions

The `state_extension` selector chooses which variables are duplicated. For one guiding-center particle, the complete dimensional convention is:

Extension	Accepted state	Splitting state	Reduced unknown	Simultaneous unknown
`physical`	$z \in \mathbb{R}^2$	$(u, v) \in \mathbb{R}^4$	$\mu \in \mathbb{R}^2$	$(u_f, v_f, \mu) \in \mathbb{R}^6$
`shared_time`	$(z, t, \kappa) \in \mathbb{R}^4$	$(u, v, t, k) \in \mathbb{R}^6$	$\mu \in \mathbb{R}^2$	$(u_f, v_f, \mu) \in \mathbb{R}^6$
`fully_extended`	$Z = (z, t, k) \in \mathbb{R}^4$	$(Z_1, Z_2) \in \mathbb{R}^8$	$\mu \in \mathbb{R}^4$	$(Z_{1f}, Z_{2f}, \mu) \in \mathbb{R}^{12}$

The literal $\mathbb{R}^2/\mathbb{R}^4/\mathbb{R}^6/\mathbb{R}^8/\mathbb{R}^{12}$ entries in the table are for one particle. For a physical run with N independent particles, the accepted, splitting, reduced, and simultaneous dimensions scale to $2N$, $4N$, $2N$, and $6N$, respectively. The shared-time and fully extended strategies currently require $N = 1$.

The accepted-state and observer-state dimensions are intentionally distinct for `shared_time`. Physical and shared-time observers receive the closed physical map $z \mapsto z_{n+1}$ in $\mathbb{R}^2$; the carried t and κ histories remain in diagnostics. Fully extended observers instead receive the accepted internal map $Z \mapsto Z_{n+1}$ in $\mathbb{R}^4$. The diagnostics fields `observer_state_dimension` and `observer_state_kind` state which convention applies.

The shared-time strategy duplicates only the physical state and shares one (t, k) pair. Its accepted conjugate variable is normalized as $\kappa = k/2$. Because the momentum update is triangular, it does not feed back into z and the physical trajectory agrees with the corresponding `physical` configuration. Its genuine $\mathbb{R}^6$ splitting state must not be confused with the temporary $\mathbb{R}^6$ nonlinear unknown of the physical simultaneous formulation.

The fully extended strategy duplicates the complete autonomous state, operates on $(Z_1, Z_2) \in \mathbb{R}^8$, and advances direct k. Its full-state projection can define a different physical map. In the simultaneous formulation, the two four-component final copies and the four-component multiplier form the $\mathbb{R}^{12}$ nonlinear workspace shown in the table. Both non-physical state extensions currently require exactly one particle so that these dimensions are literal. Every public ABBA class selects these strategies through `state_extension`; no separate extension method class is exposed.